

A $5^1 2^{30}$ -CURD ON 65 VARIETIES, COMPLETING THE $5^1 2^b$ -CURD SPECTRUM

ABSTRACT. Cordova, Diesl, Roth and Trenk proved that a class-uniformly resolvable design with partition $5^1 2^{(n-5)/2}$ can exist only for $n \in \{5, 25, 65\}$, and wrote of the last of those three orders: “*We do not know if a $5^1 2^{30}$ -CURD exists.*” It does. We exhibit one, as four base parallel classes on $\mathbb{Z}_{13} \times \{0, 1, 2, 3, 4\}$ developed by a single order-13 permutation, and the whole of $\lambda = 1$ reduces to the statement that the 160 pair orbits the four classes meet are pairwise distinct. One quarter of those labels — the forty carried by the four large blocks, which are four rows of the multiplication table of $\mathbb{Z}_{13}$ read as a difference matrix — is distinct for a reason needing no computation. Together with the two orders the source itself settles, this completes the spectrum: a $5^1 2^{(n-5)/2}$ -CURD exists if and only if $n \in \{5, 25, 65\}$. The object is printed in full, so no code is needed to check it.

1. THE OPEN CELL

Following the source [1], a *pairwise balanced (n, K, λ) -design* is a pair $(X, \mathcal{B})$ with $|X| = n$, every block size in K , and every pair of *varieties* in exactly λ blocks. Quoting [1] verbatim,

“It is resolvable if $\mathcal{B}$ can be partitioned into sets (called parallel classes) so that each parallel class contains each member of X exactly once. Furthermore, it is a Class-Uniformly Resolvable Design (CURD) if in addition, each parallel class has the same block sizes.”

The notation $m^1 2^b$ names the common block-size multiset: one block of size m and b blocks of size 2, so $n = m + 2b$. Throughout $\lambda = 1$, as in [1]; thus a $5^1 2^{30}$ -CURD on $n = 65$ varieties is an exact decomposition of the edge set of K_{65} into 52 parallel classes, each one 5-subset together with thirty disjoint pairs, with nothing doubled and nothing missed.

The target is not a conjecture but a declared gap inside a numbered theorem. Theorem 2.9 of [1] reads, verbatim,

“If $(X, \mathcal{B})$ is a CURD with partition $5^1 2^{(\frac{n-5}{2})}$ then $n = 5, 25$ or 65.”

and a few lines further on, “We do not know if a $5^1 2^{30}$ -CURD exists.” Statement numbers quoted here are those of the arXiv e-print of [1], the only full text available to us.

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Writing $m = 5$, $n = 65$ and $\lambda = 1$, the source's necessary conditions leave the cell no arithmetic slack. Theorem 2.1 of [1] gives the number of parallel classes

$$w = \frac{n(n-1)}{m^2 - 2m + n} = \frac{4160}{80} = 52,$$

Theorem 2.2 gives the number of 5-blocks through a variety

$$t = \frac{n - w - 1}{m - 2} = \frac{12}{3} = 4,$$

and Theorem 2.3 requires $mw = tn$, which holds: $5 \cdot 52 = 4 \cdot 65 = 260$. The census then closes exactly: 52 blocks of size 5 and $52 \cdot 30 = 1560$ blocks of size 2, so 1612 blocks, covering $52 \cdot ((\binom{5}{2} + 30)) = 52 \cdot 40 = 2080 = \binom{65}{2}$ pairs. Of the three orders Theorem 2.9 admits, $n = 5$ is the one-block design of Example 1.2 of [1] and $n = 25$ is built there in Theorem 4.7 and Corollary 4.11 at $q = 5$. Only $n = 65$ was undecided.

2. THE DESIGN

Theorem 1. *A $5^1 2^{30}$ -CURD on 65 varieties exists.*

Let $X = \mathbb{Z}_{13} \times \{0, 1, 2, 3, 4\}$, written c_x for the variety in column c with $\mathbb{Z}_{13}$ -coordinate x , and identified with $\{0, 1, \dots, 64\}$ by $u = 13c + x$. Let

$$\sigma: c_x \mapsto c_{x+1 \bmod 13}$$

be the order-13 permutation adding 1 inside every column. The design is the family of 52 parallel classes

$$\sigma^s(P_k), \quad k = 1, 2, 3, 4, \quad s = 0, 1, \dots, 12,$$

where the four base classes $P_1, \dots, P_4$ are these. Each line gives one block: 5-BLOCK the 5-subset, and each PAIRS line six of the thirty 2-subsets, written a|b.

```
P1 5-BLOCK 0_0 1_0 2_0 3_0 4_0
P1 PAIRS   0_1|0_8 0_2|0_12 0_3|4_12 0_4|1_10 0_5|3_2 0_6|3_4
P1 PAIRS   0_7|0_9 0_10|3_12 0_11|2_9 1_1|1_4 1_2|1_9 1_3|1_8
P1 PAIRS   1_5|3_8 1_6|2_1 1_7|3_3 1_11|2_7 1_12|3_11 2_2|2_12
P1 PAIRS   2_3|2_4 2_5|4_10 2_6|4_4 2_8|2_10 2_11|4_8 3_1|3_9
P1 PAIRS   3_5|4_3 3_6|3_7 3_10|4_7 4_1|4_2 4_5|4_11 4_6|4_9
P2 5-BLOCK 0_0 1_1 2_2 3_3 4_4
P2 PAIRS   0_1|1_6 0_2|4_9 0_3|2_6 0_4|3_8 0_5|1_12 0_6|2_7
P2 PAIRS   0_7|3_2 0_8|4_0 0_9|0_10 0_11|4_1 0_12|3_4 1_0|2_4
P2 PAIRS   1_2|3_10 1_3|4_2 1_4|2_3 1_5|1_9 1_7|2_0 1_8|2_5
P2 PAIRS   1_10|3_7 1_11|4_12 2_1|2_9 2_8|4_7 2_10|3_1 2_11|3_5
P2 PAIRS   2_12|3_11 3_0|3_9 3_6|4_11 3_12|4_5 4_3|4_8 4_6|4_10
P3 5-BLOCK 0_0 1_2 2_4 3_6 4_8
P3 PAIRS   0_1|0_10 0_2|1_6 0_3|4_1 0_4|0_9 0_5|2_10 0_6|1_4
P3 PAIRS   0_7|3_1 0_8|4_10 0_11|2_6 0_12|1_9 1_0|2_7 1_1|3_12
P3 PAIRS   1_3|4_0 1_5|2_3 1_7|4_11 1_8|3_0 1_10|3_4 1_11|4_3
P3 PAIRS   1_12|4_7 2_0|3_8 2_1|3_11 2_2|3_7 2_5|2_12 2_8|4_9
P3 PAIRS   2_9|3_2 2_11|4_6 3_3|3_10 3_5|4_12 3_9|4_5 4_2|4_4
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P4 5-BLOCK 0_0 1_3 2_6 3_9 4_12
 P4 PAIRS 0_1|1_10 0_2|2_1 0_3|2_10 0_4|4_1 0_5|2_2 0_6|4_7
 P4 PAIRS 0_7|3_8 0_8|3_7 0_9|2_5 0_10|1_5 0_11|4_4 0_12|1_11
 P4 PAIRS 1_0|1_1 1_2|2_7 1_4|4_11 1_6|1_8 1_7|4_9 1_9|3_10
 P4 PAIRS 1_12|4_10 2_0|2_9 2_3|4_6 2_4|3_0 2_8|3_6 2_11|4_5
 P4 PAIRS 2_12|4_8 3_1|3_11 3_2|3_4 3_3|4_2 3_5|4_0 3_12|4_3

The four 5-blocks are not arbitrary: for $k = 1, 2, 3, 4$,

$$(1) \quad B_k = \{ c_{(k-1)c \bmod 13} : c = 0, 1, 2, 3, 4 \},$$

which one reads off the printed lines. Thus the array $a(k, c) = (k-1)c \bmod 13$ is the 4×5 submatrix of the multiplication table of $\mathbb{Z}_{13}$ on rows $0, \dots, 3$ and columns $0, \dots, 4$ — a piece of the $(13, 13; 1)$ -difference matrix.

Pair orbits. Label the σ -orbit of a pair as follows. Since σ preserves columns and translates $\mathbb{Z}_{13}$, the orbit of $\{c_x, c_y\}$ with $x \neq y$ is determined by (S, c, d) with

$$d = \min(|x - y|, 13 - |x - y|) \in \{1, \dots, 6\},$$

and the orbit of $\{c_x, c'_y\}$ with $c < c'$ by (C, c, c', d) with $d = (y - x) \bmod 13$. Conversely each such label is realised, so the number of orbits is

$$5 \cdot 6 + \binom{5}{2} \cdot 13 = 30 + 130 = 160.$$

Lemma 2. *Every σ -orbit of a pair has size exactly 13, and consequently $160 \cdot 13 = 2080 = \binom{65}{2}$.*

Proof. σ has prime order 13, so an orbit has size 1 or 13, and size 1 would mean $\{c_x, c'_y\} = \{c_{x+1}, c'_{y+1}\}$. If $c \neq c'$ this forces $x = x+1$; if $c = c'$ it forces either $x = x+1$ or $(y = x+1 \text{ and } x = y+1)$, whence $2 \equiv 0 \pmod{13}$. Both are impossible. $\square$

Proof of Theorem 1. Each printed P_k is a partition of X into one 5-set and thirty pairs: the $5+60 = 65$ symbols listed on its six lines are $0_0, \dots, 4_{12}$, each exactly once. Since σ permutes X , every $\sigma^s(P_k)$ is again such a partition, so the 52 classes are resolutions of X with the constant block-size multiset $5^1 2^{30}$, giving $52 \cdot 31 = 1612$ blocks. The design is resolvable and class-uniform by construction; only $\lambda = 1$ remains.

Each class contributes $\binom{5}{2} + 30 = 40$ pairs, so the four base classes contribute $4 \cdot 40 = 160$ pairs, and the 52 classes contribute $52 \cdot 40 = 2080$ pairs counted with multiplicity, which is exactly $\binom{65}{2}$. Hence $\lambda = 1$ holds if and only if no pair is repeated, and since a class and its σ -translates cover a full orbit as soon as they cover one member of it, this happens if and only if

the 160 orbit labels of the 160 pairs of $P_1 \cup P_2 \cup P_3 \cup P_4$ are pairwise distinct.

They are. Reading the printed lines, the 30 same-column labels (S, c, d) each occur once and the 130 cross-column labels (C, c, c', d) each occur once, for 160 distinct labels; by Lemma 2 their orbits partition the 2080 pairs. $\square$

One quarter of that final tally is forced by (1) and needs no computation. Each B_k meets every column once, so it contributes no same-column label, and on the column pair (c, c') with $c < c'$ it contributes the cross-label

$$d_k = (k-1)c' - (k-1)c = (k-1)\delta \pmod{13}, \quad \delta = c' - c \in \{1, 2, 3, 4\}.$$

As k runs over 1, 2, 3, 4 these are $0, \delta, 2\delta, 3\delta$, pairwise distinct modulo the prime 13. So the four 5-blocks alone already use 40 distinct labels, four on each of the ten column pairs; the content of the construction is that the remaining 120 pairs cover the remaining 120 orbits — all 30 same-column ones and the $10 \cdot (13-4) = 90$ surviving cross-column ones — and $120 \cdot 13 = 1560 = 52 \cdot 30$ is the number of 2-blocks, so again there is no slack.

Remark. The 5-blocks are more structured than $\lambda = 1$ demands. By (1), the variety c_x lies in the 5-block of $\sigma^s(P_k)$ if and only if $s \equiv x - (k-1)c \pmod{13}$, one solution in $\mathbb{Z}_{13}$; so for each fixed k the thirteen 5-blocks $\sigma^s(B_k)$ themselves form a parallel class of X . The 52 5-blocks are therefore distinct, they meet pairwise in at most one variety, and each variety lies in exactly $t = 4$ of them — the four families, one apiece.

Corollary 3. *A $5^1 2^{(n-5)/2}$ -CURD exists if and only if $n \in \{5, 25, 65\}$.*

Proof. Necessity is Theorem 2.9 of [1], which we quote and do not reprove. Sufficiency: $n = 5$ is Example 1.2 of [1], $n = 25$ is its Theorem 4.7 with Corollary 4.11 at $q = 5$, and $n = 65$ is Theorem 1 above. $\square$

3. A NEAR MISS

A published theorem prints the partition $5^1 2^{30}$ on 65 points and is a different object, so we name it. Theorem 3.5 of Danziger and Stevens [2] states, verbatim:

“For q a prime power, $2 \leq k \leq q$, $0 \leq m \leq q/(k-1)$, there exists a $\text{CURGDD}_{q-m(k-2)}$ of type q^k with partition $2^{mk(k-1)/2} k^{q-m(k-1)}$, satisfying the conditions of Theorem 3.3.”

At $(q, k, m) = (13, 5, 3)$ — legal, since $3 \leq 13/4$ and $5 \leq 13$ — this reads: type 13^5 , that is 65 points, with partition $2^{30} 5^1$. It is not the object of Theorem 1, for two reasons that are pure arithmetic.

- (1) It is *group divisible* of type 13^5 . The $5 \binom{13}{2} = 390$ within-group pairs are covered 0 times, and only $\binom{5}{2} \cdot 13^2 = 1690$ of the 2080 pairs are covered at all, whereas a CURD covers all 2080.
- (2) Its index is $\lambda = q - m(k-2) = 4$, and $4 \cdot 1690/40 = 169$ resolution classes, not $\lambda = 1$ with 52.

Nor can the family be specialised to repair this: $\lambda = 1$ requires $q - m(k - 2) = 1$ while “exactly one k -block per class” requires $q - m(k - 1) = 1$, and subtracting gives $m = 0$, hence $q = 1$. The only $\lambda = 1$ CURD conclusion of [2] is its Corollary 3.8, which lands on $q^2 + q + 1$ points; since $7^2 + 7 + 1 = 57$ and $8^2 + 8 + 1 = 73$, the value 65 is not of that form.

4. SCOPE, AND VERIFICATION

This is a construction, not a classification. Uniqueness, enumeration and the number of solutions at $n = 65$ are untouched, and no non-existence assertion is made anywhere in this note. Two of the three members of the spectrum in Corollary 3 are the source’s, not ours: we reproduce no object at $n = 5$ or $n = 25$ and re-derive only their parameter arithmetic, and the necessity half of that corollary is quoted from [1] and was not reproved. The source states a second undecided cell, a $4^1 2^{12}$ -CURD on $n = 28$; nothing above bears on it in either direction. No resolvable $(65, 5, 1)$ -design is claimed or used.

The object is printed in full above and the argument of Section 2 is finite and short, so a reader can check Theorem 1 from the page: confirm that each P_k lists the 65 varieties once, read off the 160 orbit labels, and confirm they are distinct. The accompanying program `verify.py` performs that check mechanically, in exact integer arithmetic, from a copy of the printed block held inside it; it also develops all 52 classes and re-derives the pair-multiplicity multiset $\{1:2080\}$ directly, without the orbit argument, though both routes read that one parsed copy through the same implementation. It reports 53 checks, all passing, and states in its own output what it does not cover. The recorded run `verify.output.txt` carries the SHA-256 of `verify.py`, and the program prints the SHA-256 of the object string it read, so the bytes checked can be matched against the bytes printed above.

REFERENCES

- [1] R. Cordova, A. Diesl, H. Roth, and A. Trenk, *Class-uniformly resolvable designs with all but one block having size two*, arXiv:2606.27519v1, 2026. arXiv:2606.27519.
- [2] P. Danziger and B. Stevens, *Class-uniformly resolvable group divisible structures I: resolvable group divisible designs*, Electron. J. Combin. **11** (2004), no. 1, Research Paper 23. doi:10.37236/1776.